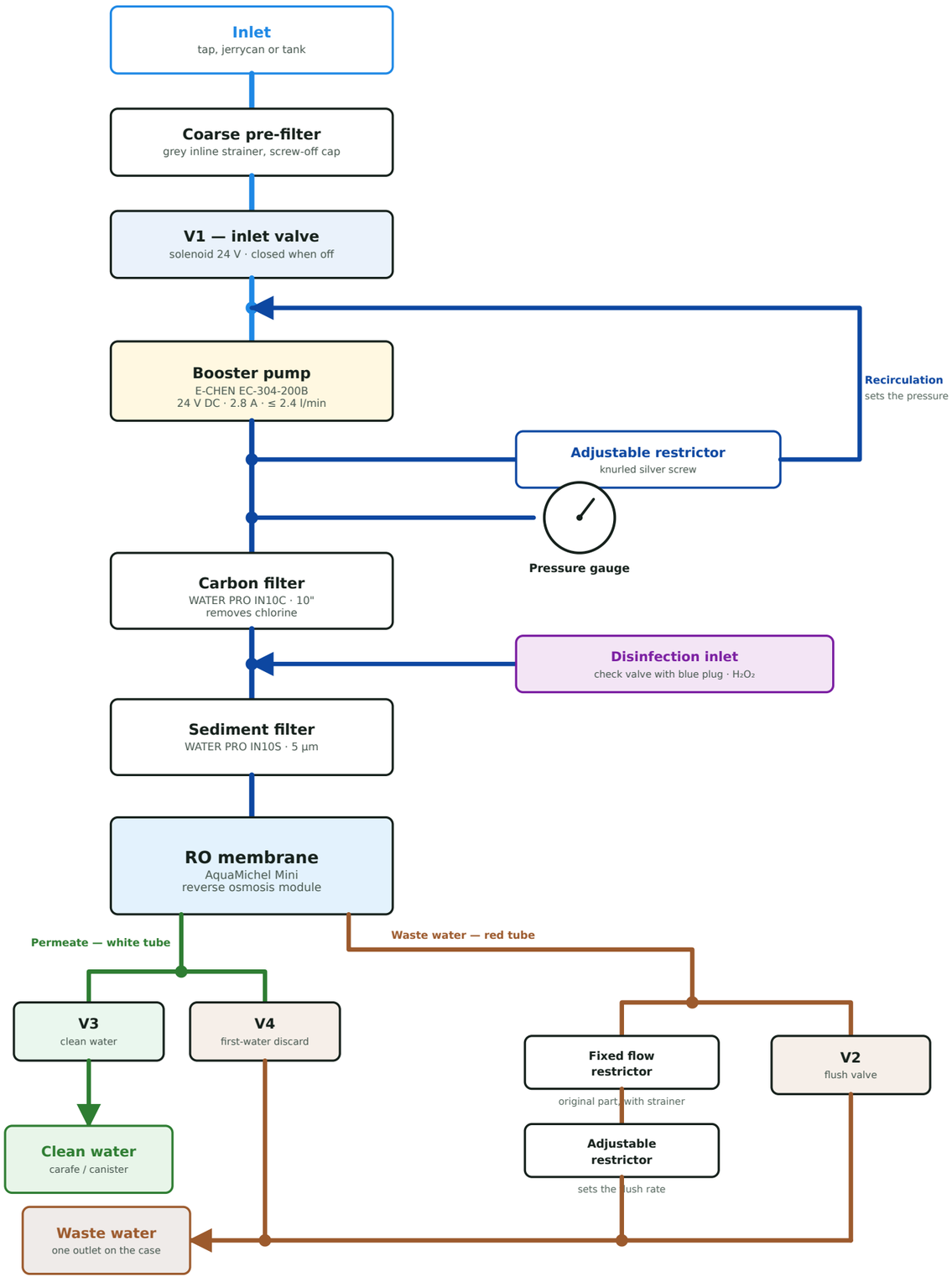
Hydraulics — every water-carrying part

Reverse osmosis system in a carry case · AquaMichel Mini with booster pump and four solenoid valves



Legend

- Supply, unpressurised (before the pump)
- Pressure side (pump → membrane)
- Permeate / clean water
- Waste water / flush water
- Tee / branch

Valve positions per mode (from main.py)

Mode	V1	V2	V3	V4	Pump
Idle	closed	closed	closed	closed	OFF
Flush	**open**	**open**	closed	closed	**ON**
Discard	**open**	closed	closed	**open**	**ON**
Filter	**open**	closed	**open**	closed	**ON**
Pressure bleed	closed	**open**	closed	closed	OFF

Notes for rebuilding

Operating modes

Filter: V1 + V3 open. Waste water leaves only through the restrictors, which therefore set the recovery rate.
Flush: V1 + V2 open. V2 bypasses both restrictors, giving a high cross-flow that sweeps the membrane clean.
Discard: V1 + V4 open. The first permeate after a standstill goes to the waste outlet instead of into the glass.
Pressure bleed: V2 alone open, pump off — residual pressure escapes through the waste line.

Compared with the original build manual

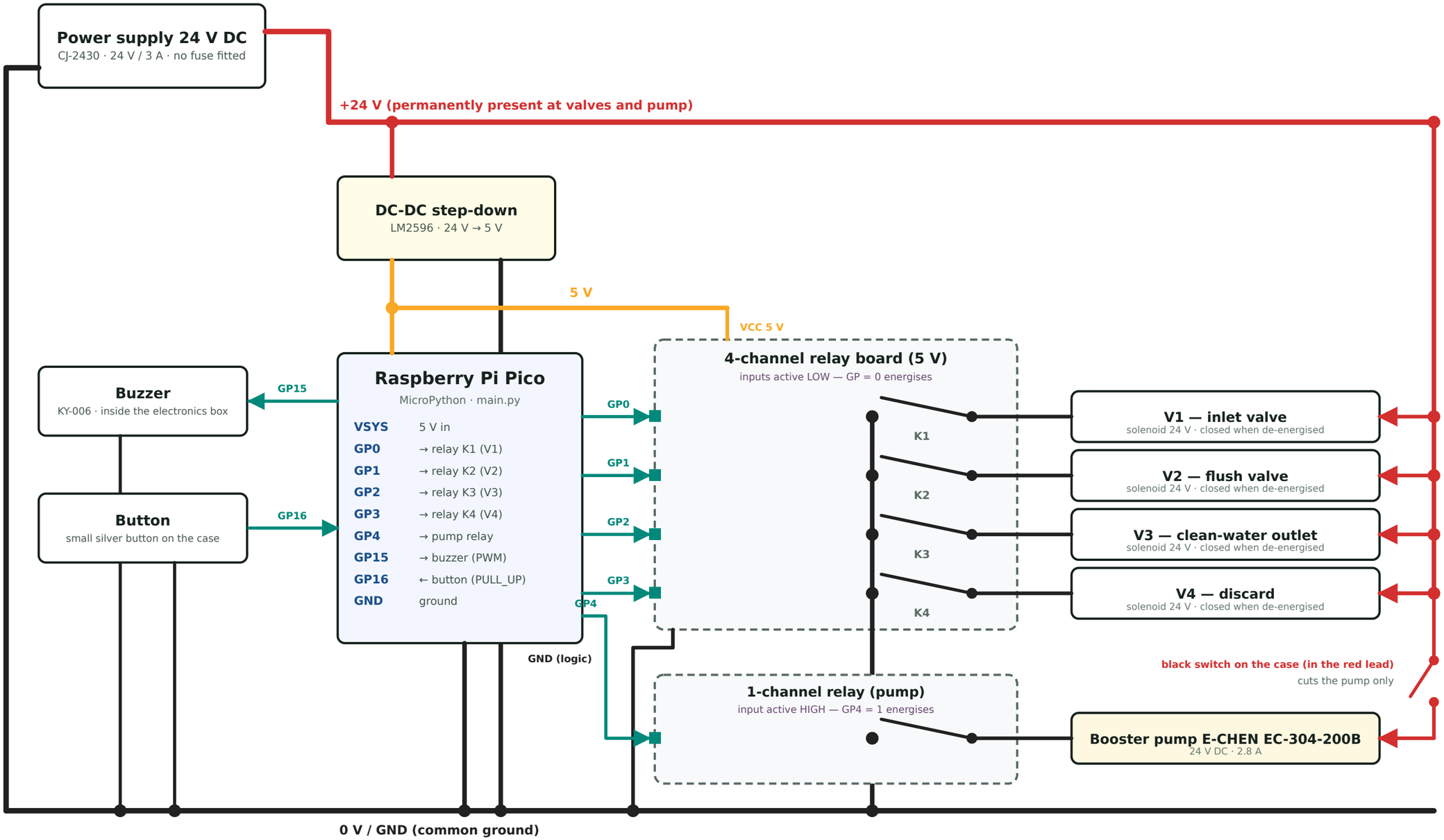
V2 replaces the blue ball valve of the flush-water system.
The order carbon → sediment → membrane is the one the AquaMichel Mini manual prescribes, not a mistake.
The fixed restrictor carries a strainer that keeps scale out of the adjustable one — keep it in that order.

Tube colours in this build

white = permeate from the membrane, red = waste water from the membrane, blue = supply and pre-filter run. Red is also used on the pressure side, so do not read the colours as a strict code.

Electrics — controller, valves and pump

Raspberry Pi Pico · 24 V DC · the relays switch the negative line



+24 V (pump, valves) · 0 V / switched ground · 5 V (logic) · GPIO control signal

- Switching principle: +24 V sits permanently on every load. The relay breaks the negative line — at rest the contact is open, so the load is dead.
- Relay polarity: the 4-channel board energises on LOW, the pump relay on HIGH. main.py already matches this (_set_valves inverts, set_pump does not).
- Tie all grounds together: supply −, step-down −, Pico GND, both relay boards. Without a common ground the relays switch erratically.
- Recommended when rebuilding: a 5 A slow-blow fuse in the +24 V line and a 5 A supply — the pump alone draws 2.8 A of the 3 A available.